A LOCALLY QUASI-FINITE SEPARATED MORPHISM OF ALGEBRAIC SPACES IS REPRESENTABLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is a one-step deduction from the proposition plus base change stability; the author writes that the claim is immediate from these.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- A locally quasi-finite separated morphism to a representable base has representable source. Proposition proposition-locally-quasi-finite-separated-over-scheme (10806-10817), used at proof lines 10936-10938. Is the criterion applied after base change.
- Stability of local quasi-finiteness and separatedness under base change. Lemmas lemma-base-change-quasi-finite and lemma-base-change-separated, used at proof lines 10939-10942. Makes the base change hypotheses available.
- Representability tested after base change by representable objects. used at proof lines 10936-10943. Is the definition of representability used here.

Source reference. Original lemma 03XL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5152–5152. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03KL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4681–4681. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original proposition 03XX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5164–5164. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. If f is locally quasi-finite and separated, then f is representable.*

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. This is immediate from Proposition 03XX and the fact that being locally quasi-finite and separated is preserved under any base change, see Lemmas 03XL and 03KL. □